Applied Cryptography Sample Test II

Name and Surname: ____

1. What is the polynomial representation of:

$$0xCA = 11001010$$

A. $x^7 + x^6 + x^3 + x^2 + x$
B. $x^7 + x^6 + x^3 + x$
C. $x^7 + x^4 + x^3 + x$
D. $x^6 + x^3 + x$

2. What is the hexadecimal value of:

$$x^7 + x^4 + x^3 + x + 1$$

A. 0x93
B. 0x9B
C. 0x8B
D. 0xBB

3. Compute:

$$0xA7 \oplus 0x3C$$

A. 0x83
B. 0x9B
C. 0xAB
D. 0x97

4. Compute:

$$0xFF \oplus 0x0F$$

A. 0xF1
B. 0xF0
C. 0x0F
D. 0xFF

5. In GF(2), what is:

$$(x+1)(x+1)$$

A. $x^2 + 2x + 1$
B. $x^2 + x + 1$
C. $x^2 + 1$
D. $x + 1$

6. In AES arithmetic:

$$x^8 =$$

A. $x^7 + x^4 + x + 1$
B. $x^4 + x^3 + x + 1$
C. $x^8 + 1$
D. $x^5 + x^4 + x$

7. What is:

$$x^9 m(x)$$

A. $x^4 + x^3 + x + 1$
B. $x^5 + x^4 + x^2 + x$
C. $x^5 + x^3 + x + 1$
D. $x^6 + x^2 + 1$

8. AES operates on elements of:

A. $GF(256)$
B. $GF(8)$
C. $GF(2^8)$
D. $GF(1024)$

9. Addition in $GF(2^8)$ is performed using:

A. AND
B. OR
C. XOR
D. NOT

10. Why must the AES polynomial be irreducible?

A. To reduce memory usage
B. To speed up XOR
C. To ensure a finite field structure
D. To shorten keys

11. SHA-256 always produces:

A. 128 bits
B. 512 bits
C. 256 bits
D. Variable length output

12. A SHA-256 hash contains:

A. 16 bytes
B. 32 bytes
C. 64 bytes
D. 128 bytes

13. A SHA-256 hash written in hexadecimal contains:

A. 32 characters
B. 64 characters
C. 128 characters
D. 256 characters

14. SHA-256 processes data in blocks of:

A. 128 bits
B. 256 bits
C. 512 bits
D. 1024 bits

15. How many rounds are performed for each SHA-256 block?

A. 16
B. 32
C. 64
D. 128

16. What property ensures that the same input always produces the same hash?

A. Collision resistance
B. Determinism
C. Avalanche effect
D. Encryption

17. What is the avalanche effect?

    A. Faster hashing
    B. Smaller hashes
    C. Small input changes produce very different outputs
    D. Automatic encryption

18. Which class should be used for cryptographic random values?

    A. Random
    B. SecureRandom
    C. Math
    D. ThreadLocalRandom

19. Why is a salt added before hashing?

    A. To shorten passwords
    B. To compress data
    C. To make identical passwords produce different hashes
    D. To encrypt files

20. Why is AES preferred over DES?

    A. AES uses smaller keys
    B. DES is faster
    C. AES supports larger keys and is more secure
    D. AES does not require keys